GlycoTwin • System Architecture

Type 2 Diabetes Physiological Digital Twin & Predictive Decision Support Engine

Happiest Health Challenge 2026

`Multi-Modal Deep Fusion (GRU + MLP)`

1. INGESTION SOURCES

Synthetic Cohort — 10,000 Pts

- Nathan et al. 2008 calibration ($r = 0.98$)
- Gamma postprandial curves (30–60m peak)
- Full hypnograms & circadian HR / HRV
- Realistic steps (2,000–12,000/day)

`src/data_generator/`

Wearable Adapters — Real Devices

- Abbott FreeStyle Libre (CSV export)
- Dexcom Clarity CGM format
- Apple Health export XML/CSV
- Simulated AI-READI, CGMacros, Shanghai

`src/data_real/`

2. IN-MEMORY VALIDATION

Validator & Sanitizer — In-Memory

- Auto-unit conversion (mmol/L → mg/dL)
- 5-minute grid uniform resampling
- Cubic spline imputation (<30m gaps)
- Strict 24h duration gate (≥288 rows)

`src/ingestion/validator.py`

Leakage Auditing — Zero-Leakage

- Disjoint patient splits (Train/Val/Test)
- Temporal causality: 24h lookback strictly precedes 2h forecast horizon
- Target: $G \ge 180$ within 2h given $G < 180$

`src/features/builder.py`

3. MULTIMODAL FUSION

Dynamic Sensor GRU — Lookback

- Input: [288 timesteps × 6 channels]
- Channels: CGM, HR, HRV, steps, sleep, carbs
- 2-Layer BiGRU (64 units, dropout=0.2)
- Temporal self-attention pooling → 64-d

`Sensor Branch: 0.9210 AUROC`

Static EHR MLP — Demographics

- Input: 16 clinical EHR biomarkers
- Age, Sex, BMI, HbA1c, Fasting Glu, BP, Meds
- 2-Layer MLP (16 → 64 → 32, LayerNorm)
- Cross-modal concatenation head (96-d)

`Fusion Model: 0.9008 AUROC`

4. DIGITAL TWIN ENGINE

PatientTwin State — Encapsulation

- Real-time sensor trace & rolling stats
- Calibrated baseline physiological profile
- Active 2h spike probability & risk tier
- Status: Optimal, Watch, or Hyperglycemic

`src/twin/state.py`

What-If Simulator — Counterfactual

- Post-meal brisk walk (GLUT4 translocation)
- Carbohydrate challenge / snack omission
- Metformin adherence & sleep hygiene
- Instantaneous projected risk delta (↓ pts)

`src/twin/what_if.py`

5. CLINICIAN INTERFACE

Decision Dashboard — Streamlit

- 10-second Executive Summary & Result Card
- Interactive CGM spline & 2h forecast cone
- Ambulatory Glucose Profile (AGP) modal day
- Clinical Light & Dark Theme (WCAG AA)

`app/streamlit_app.py`

Explainability & Audit — SHAP Attrib

- Signed horizontal feature impacts (% risk)
- Scientific Model Report & fair baselines
- F1/Precision/Recall vs threshold curve
- In-memory private drag-and-drop upload

`app/ui_theme.py`